MILITARY SPECIFICATION

CONNECTOR, PLUG, ELECTRICAL
CONNECTOR, RECEPTACLE, ELECTRICAL
(HEAVY DUTY, WATERPROOF)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers heavy duty, waterproof, electrical connector plugs and electrical receptacles (see 6.1).

1.2 Classification. Connector plugs and receptacles shall be of the following types, styles, classes and sizes as specified (see 6.2).

Type I	— Straight shell
Type II	— Angle shell
Style P	— Pin contact
Style S	— Socket contact
Class 1	— Plug with wire mesh cable grip
Class 2	— Plug with spring cable grip

1.2.1 *Size.* The connector size shall be designated by the thread size of the shell or the coupling nut, whichever is applicable and shall be one of the following:

1.906	—	16 NS
2.500	—	14 NS
3.750	—	12 UN

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP–B–566	— Boxes, Folding, Paperboard.
PPP–B–585	— Boxes, Wood, Wirebound.
PPP–B–591	— Boxes, Fiberboard, Wood-Cleated.
PPP–B–601	— Boxes, Wood, Cleated-Plywood.
PPP–B–621	— Boxes, Wood, Nailed and Lock-Corner.
PPP–B–636	— Boxes, Fiber.
PPP–B–676	— Boxes, Set-Up Paperboard.
PPP–T–76	— Tape, Pressure Sensitive, Adhesive, Paper, Water Resistant for Carton Sealing.

FSC 5935

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MIL–P–116 — Preservation, Methods of.

MIL–B–130 — Barrier-Material, Paper, Noncorrosive.

MIL–C–13777 — Cable, Special Purpose, Electrical.

MIL–I–45208 — Inspection Requirements General Specification For.

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

MIL–STD–130 — Identification Marking of U.S. Military Property.

MIL–STD–171 — Preparation, Painting and Finishing of Metal and Wood Surfaces.

MIL–STD–194 — Systems for Painting and Finishing Fire-Control Material.

MIL–STD–202 — Test Methods for Electronic and Electrical Component Parts.

(For applicable MS Military Standards, see Supplement 1.)

(Copies of specifications, standards, drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitations for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, One Park Avenue at 33d St., New York 16, N. Y.)

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification No. A–13 and No. 13.

(Application for copies of the above publications should be addressed to the American Trucking Associations, Inc., Traffic Department, 1424 16th Street, N. W., Washington, D. C.)

U. S. POST OFFICE DEPARTMENT

Parcel Post Regulations.

(Application for copies should be obtained from any first class U. S. Post Office.)

3. REQUIREMENTS

3.1 Design and construction. The design and construction of the connectors shall be in accordance with the MS standards listed in Supplement 1 which form a part of this specification.

3.2 Impregnation. Connector shells and covers shall be impregnated in accordance with Standard MIL–STD–194. The impregnated castings shall be capable of withstanding an air pressure of 8 pounds per square inch without signs of leakage.

3.3 Finish.

3.3.1 *Tin-plating.* Contacts shall be tin-plated in accordance with finish 1.8 of Standard MIL–STD–171 and shall show no evidence of corrosion when tested as specified in 4.8.5.

3.3.2 *Painting.* All exterior surfaces except threads and surfaces designated as unfinished on drawings shall be olive drab in

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accordance with System 21.5 of Standard MIL–STD–171. Where colors other than olive drab are required they shall be specified by the procuring activity and shall be applied over the standard finish (see 6.2).

3.4 Polarization. Connector plugs and receptacles shall be polarized with the polarization being accomplished before the pin and sockets are engaged.

3.5 Service rating. The voltage and current ratings for connector plugs and receptacles shall be as specified in table I.

TABLE I. *Service Ratings of Plugs and Receptacles*

No. of poles	Current amperes (max)	Wire size range	AC [1] voltage	DC voltage
38	30	12 — 18	150	200
28	30	12 — 18	150	200
	Small 30 Contact	14 — 18		
21	Large 45 Contact	8 — 12	150	200
19	30	14 — 18	150	200
4	60	8 — 10	275	500
3	45	8 — 10	150	200
3	200	0 — 1	275	500

[1] Voltage ratings are the maximum operating voltages between adjacent contacts or between the shell and nearest contact and represent the limiting operating voltage at sea level.

3.6 Insulation resistance. The insulation resistance shall be a minimum of 5000 megohms when a potential of approximately 500 volts is applied between any pair of contacts and between the shell and any contact.

3.7 High potential. Connector plugs and receptacles shall show no evidence of breakdown when the voltage indicated in Table II is applied between the two closest contacts and between the shell and the contacts closest to the shell for 1 minute.

TABLE II. *Voltage for High Potential Test*

Connector rating ac volts (rms)	Test voltage ac volts (rms)
150	1300
275	1550

3.8 Waterproofness. Mated connectors when assembled with proper cable in accordance with Specification MIL–C–13777 and tested as specified in 4.8.2 shall show no evidence of water inside the plugs. The plugs shall withstand a potential of 500 volts ac (rms) plus twice the service voltage rating between adjacent contacts and between the shell and the closest contacts for a period of 5 minutes.

3.9 Resistance of contacts. The resistance of the contacts shall be such, that the potential drop at the specified current shall not be greater than as specified in table III.

TABLE III. *Contact Potential Drop*

No. of poles	Test current amperes	Potential drop millivolts (max)
3	200	10
3	45	15
4	45	15
19	15	26
	Small 15 Contacts	26
21	Large 35 Contacts	15
28	15	26
38	15	26

3.10 Marking. When specified in the contract all parts shall have the part number molded, stamped, etched or engraved on the surface of the part in accordance with Standard MIL–STD–130 (see 6.2).

3.11 Workmanship. The connectors shall be manufactured so as to be uniform in quality and free from sharp edges, burs and other defects that would impair the function of the item.

4. QUALITY ASSURANCE PROVISIONS

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4.1 General provisions. The quality assurance provisions of referenced documents, to the extent specified herein, together with provisions of this section are applicable to inspection performed by the contractor and for Government verification.

4.1.1 The contractor shall provide an inspection system in accordance with Specification MIL–I–45208.

4.2 Lot formation. Inspection lot sizes and lot formation shall be in accordance with Standard MIL–STD–105.

4.2.1 Defective units and lots shall be processed as specified in Standard MIL–STD–105.

4.3 Inspection of components. All positional and functional characteristics shown on the drawings which indicate precision requirements and have a bearing on functional accuracy and performance of the connectors shall be inspected in accordance with the written inspection system required by Specification MIL–I–45208. These characteristics are defined as drawing requirements affecting identification marking and performance requirements specified in notes on the drawings and in separate performance specifications referenced on the drawings.

4.3.1 *Sampling inspection.* Sampling inspection of components may be used in lieu of 100 percent inspection and shall be in accordance with table IV.

TABLE IV. *Sampling Plan*

Lot size	*Sample size*	*Accept lot*	*Reject lot*
0–25	Use 100% Inspection		
26–35	20	0	1
36–50	25	0	1
51–150	30	0	1
151–Over	30	1	1

4.3.2 *Disposition of nonconforming product.* Acceptance and rejection numbers are expressed in terms of defective units; one or more defects on any unit shall constitute a defective unit. Disposition of nonconforming components shall be in accordance with Standard MIL–STD–105.

4.4 Material. Control of material shall be in accordance with test data requirements of Specification MIL–I–45208.

4.5 Inspection equipment.

4.5.1 *Contractor's provisions.* The contractor shall supply, calibrate, maintain and use inspection equipment as required by Specification MIL–I–45208.

4.6 Tests.

4.6.1 *Classification of defects.* The classification of defects in tables V, VI, VII, and VIII shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots. The Government reserves the right to inspect for any or all requirements and to reject individual nonconforming items.

4.6.1.1 Samples selected for acceptance tests listed under tables VI and VII shall include those samples selected for acceptance testing specified in table V.

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TABLE V. *Classification of Defects*

Use Inspection Level L8, and Sampling Plan Table IV–A
of Standard MIL–STD–105

Critical—None Defined *Major*	*AQL 6.5%*	*Requirement paragraph*	*QA paragraph*	*Test method*
101	Waterproofness	3.8	4.8.2	SME and Visual
102	Resistance of contacts	3.9	4.8.3	SME
103	High potential	3.7	4.8.4	SME
104	Tin plating	3.3.1	4.8.5	MIL–STD–202

TABLE VI. *Classification of Defects*

Use Inspection Level II, and Sampling Plan Table IV–A
of Standard MIL–STD–105

Critical—None Defined *Major—AQL 1.5%*		*Requirement paragraph*	*QA paragraph*	*Test method*
101	Design and construction	3.1	4.8.1	Visual
102	Polarization	3.4	4.8.6	Visual
103	Insulation resistance	3.6	4.8.7	Method 302 MIL–STD–202
104	Markings	3.10	4.8.8	Visual
Minor—AQL 2.5%				
201	Painting	3.3.2	4.8.9	MIL–STD–171
202	Workmanship	3.11	4.8.10	Visual
Special				
301	Impregnation	3.2	...	Certification

4.7 Acceptance by certification. Certification specified in Table VI shall be executed by the contractor. The contractor shall be responsible for all inspections and tests necessary to insure that the characteristic specifying certification as the method of inspection for Government acceptance is in accordance with all related requirements contained in the contract or order.

4.7.1 The certification shall be based on either the contractor inspection tests or those conducted by other qualified laboratory or inspection organizations.

4.7.1.1 The certification shall delineate all drawing and related specification requirements being attested to and shall stipulate the results of such tests.

4.7.1.2 All certifications shall be signed by a responsible officer of the contractor.

TABLE VII. *Classification of Defects*

Use Inspection Level II, and Sampling Plan Table IV–A
of Standard MIL–STD–105

Critical—None Defined *Major—AQL 1.5%*		*Requirement paragraph*	*Test method*
101	Cleaning (Level A)	5.1.1.1	Visual
102	Unit packaging (Level A, C)	5.1.1.2, 5.1.2	MIL–P–116
103	Intermediate packaging (Level A)	5.1.1.3	Visual
104	Packing (Level A, B, C)	5.2.1, 5.2.2, 5.2.3	Visual
105	Marking	5.3	Visual
Minor—None Defined			

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TABLE VIII. *Classification of Defects*

Use Inspection Level L8, and Sampling Plan Table IV–A
of Standard MIL–STD–105

		Requirement paragraph	Test method
Critical—None Defined			
Major—AQL 6.5%			
101	Heat seal	5.1.1.2, 4.8.11	MIL–P–116
102	Quick leak	5.1.1.2, 4.8.12	MIL–P–116
Minor—None Defined			

4.8 Inspection testing.

4.8.1 *Design and construction.* Connectors shall be inspected visually and by means of standard measuring equipment for compliance with the requirements of 3.1.

4.8.2 *Waterproofness.* Connectors assembled as specified in 3.8 shall be submerged under a water pressure equal to 5 feet of water for a period of 5 days. Immediately after removal from the water the connectors shall be examined and tested for conformance with 3.8.

4.8.3 *Resistance of contacts test.* After the completion of the salt spray test the electrical resistance of the contacts shall be determined by measuring the potential drop across the contacts when subjected to the specified test current in table III. The potential drop through the contacts shall be measured with the probes perpendicular to the wires and touching both the wire and the contact with the side of the probe as shown in figure 1.

4.8.4 *High potential.* The connectors shall be tested in accordance with Standard MIL–STD–202, Method 301, with the voltage specified in table II applied between the closest contacts and between the shell and the contacts closest to the shell for a period of 1 minute.

4.8.5 *Salt spray.* Pin and socket contacts shall be subjected to the salt spray test in accordance with Standard MIL–STD–202, Method 101, for a period of 50 hours. Immediately after exposure the surfaces shall be washed with tap water and dried for 12 hours in a circulating air oven at a temperature of 38° plus or minus 3° C. (100° plus or minus 5° F.). Exposure to the salt laden atmosphere shall not cause sufficient corrosion to expose the base metal.

4.8.6 *Polarization.* All connectors shall be visually checked to determine that polarization is compatible with the MS standards.

4.8.7 *Insulation resistance.* The insulation resistance shall be measured separately between any pair of contacts and between the shell and the contact, using a potential of 500 volts for a period not exceeding 2 minutes. Measurements shall be made in accordance with Standard MIL–STD–202, Method 302.

4.8.8 *Marking.* The connector parts shall be visually inspected for compliance with Standard MIL–STD–130 and the requirements of 3.10.

4.8.9 *Painting.* The test method outlined in Standard MIL–STD–171 shall apply to determine that the connectors conform to the requirements of 3.3.2.

4.8.10 *Workmanship.* The connectors shall be visually inspected for compliance with 3.11.

4.8.11 *Heat sealing.* Test procedures outlined in Specification MIL–P–116 shall apply.

4.8.12 *Quick leak test.* Test procedures outlined in Specification MIL–P–116 shall apply.

4.8.13 *Level C shipping container.* Connec-

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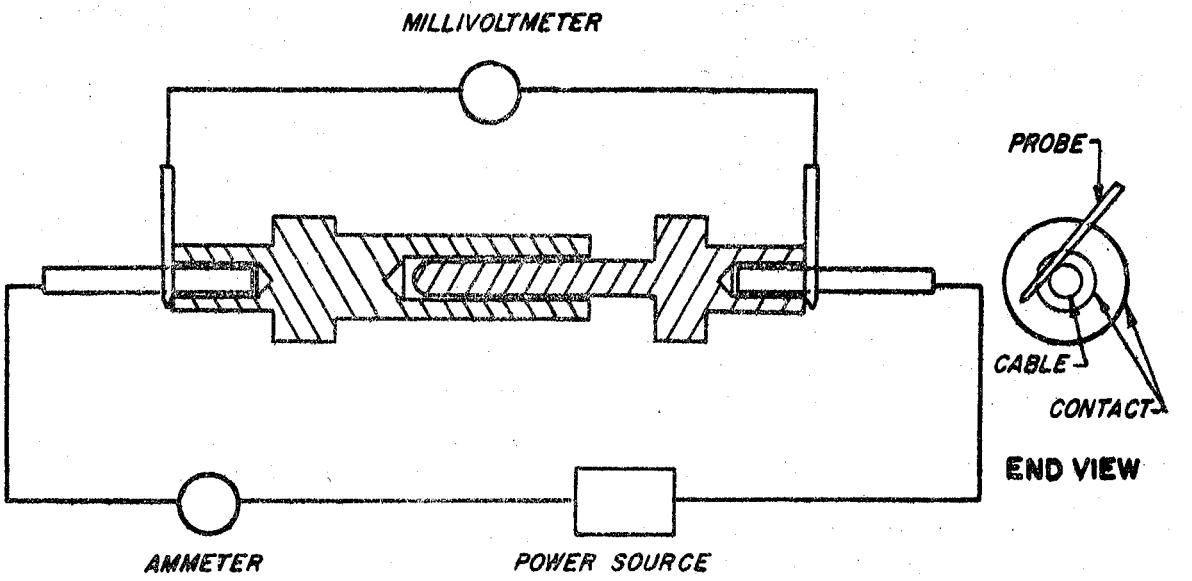


FIGURE 1. *Diagram for resistance of contacts test.*

tors packaged as specified in 5.1.1 shall be packed to the applicable regulation for the mode of transportation utilized in compliance with 5.2.3.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Cleaning.* The connectors shall be cleaned with dry compressed air to remove any dust or foreign matter.

5.1.1.2 *Unit packaging.* The connectors shall be individually wrapped with barrier material conforming to Specification MIL–B–130 and packaged in accordance with Method IA–8 of Specification MIL–P–116. The bagged items shall be inclosed in folding cartons or set-up boxes conforming to Specifications PPP–B–566 or PPP–B–676.

5.1.1.3 *Intermediate packaging.* Connectors packaged in accordance with 5.1.1.2 shall be further packaged in quantities of 5 unit packages or multiples thereof, in intermediate containers conforming to Specifications PPP–B–566, PPP–B–636 or PPP–B–676 with a weight limitation of 20 pounds. The box shall be closed by applying 2 inch width tape conforming to Specification PPP–T–76 to top and bottom seams. If bottom of the container is closed with staples or adhesive only a single piece of tape on the top seam is required.

5.1.2 *Level C.* Connectors shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 *Level A.* Connectors of the same stock number packaged as specified in 5.1.1.2 and 5.1.1.3 shall be packed in a close fitting wire-bound wood box, wood-cleated fiberboard box, wood-cleated plywood box, nailed wood box or a fiberboard box conforming to Specifications PPP–B–585, class 3; PPP–B–591 (Overseas type); PPP–B–601, style A or B; PPP–B–621, class 2; and PPP–B–636, class 2 or 3 respectively. Closure and strapping shall be in accordance with the applicable box specification. The gross weight of the boxes shall not exceed 200 pounds.

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5.2.2 *Level B.* Connectors of the same stock number packaged as specified in 5.1.1.2 and 5.1.1.3 shall be packed in a close-fitting wirebound wood box, wood-cleated fiberboard box, wood-cleated plywood box, nailed wood box or a fiberboard box conforming to Specifications PPP–B–585, class 1; PPP–B–591 (Domestic type); PPP–B–601 (Domestic type); PPP–B–621, class 1; and PPP–B–636, class 1 respectively. Closure and strapping shall be in accordance with the applicable box specification. The gross weight of the boxes shall not exceed 200 pounds.

5.2.3 *Level C.* Connectors shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules, Motor Freight Regulations or Parcel Post Regulations, whichever is applicable.

5.3 Marking. Unit packages, intermediate containers and shipping containers shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 Intended use. Electrical connectors are intended to provide an easy and convenient means for connecting and disconnecting electrical circuits in portable cable systems which are exposed to adverse weather conditions.

6.1.1 *Straight connectors.* Straight connectors are intended for use in connecting circuits where available space is unlimited.

6.1.2 *Angle connectors.* Angle connectors are intended for use where available space for the connection is limited.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) MS part number.

(c) Type, style, class and size required.

(d) Selection of applicable levels of packaging and packing required.

(e) Ratings.

(f) Color.

(g) Responsibility for supplying material for inspection tests.

6.3 Definitions.

6.3.1 *Standard measuring equipment (SME).* Standard measuring equipment is standard hand-type measuring devices which are usually stocked by commercial supply houses and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

Custodians:
Army—SigC
Navy—BuShips

Preparing activity:
Army—Ord

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